

Crowd-Driven Visual Generation with Conditional Diffusion Models

Course Project Report

Course: Generative AI: Principles and Applications, CCE IISc

Author: Nishant Kumar (nishantkro39@gmail.com)

Date: July 26, 2026

Submission Type: Course Project Report

Abstract

Turning a crowd of short, heterogeneous responses (words, emojis, doodle labels) into one coherent image is a *many-to-one* conditioning problem: the input is an unordered, noisy set, not a single prompt. We study the design choice this exposes — how to aggregate many responses into one conditioning signal — and compare four **permutation-invariant** strategies (mean-pool, cluster-centroid, Deep-Sets, attention-pooling) inside a from-scratch conditional latent diffusion model, with a conditional GAN as an alternative generator. Our central result (RQ1) is that learned aggregation (attention $\approx$ Deep-Sets) **significantly outperforms heuristic pooling** (mean $\approx$ centroid) at producing on-theme images — a difference of **0.034** in CLIP theme-fidelity (95% CI [0.031, 0.036], $p \approx 3 \times 10^{-103}$) that holds as crowd diversity rises and **reproduces across both generative backbones** (RQ2). Reaching this required diagnosing and mitigating a CLIP **modality gap** (a model trained on image embeddings responds weakly to a crowd's text embeddings) via a text-conditioned retrain. We report the negative results (two failed inference-time fixes) and limitations alongside the positive findings.

1. Introduction

Existing text-to-image models assume a single prompt authored by a single user. Yet modern interactive settings — concerts, classrooms, sporting events, online communities — increasingly gather hundreds or thousands of simultaneous audience contributions to a shared prompt ("describe how this moment feels"). Turning that collective input into one coherent image is a fundamentally different conditioning problem. The input is a set (no order), it is heterogeneous (words, emojis), it is large (hundreds of responses), and it is noisy (not everyone stays on theme). Success therefore depends less on prompt engineering and more on robust aggregation of a noisy set — the step we study here.

Concretely, the interesting question is the many-to-one conditioning step: how do we collapse a noisy crowd into a conditioning vector that steers a generator toward the crowd's dominant intent while ignoring off-theme responses?

Contributions.

1. We formulate crowd-driven generation as a many-to-one conditioning problem (§2) — mapping an unordered, noisy set of audience responses to a single conditioning vector — and study permutation-invariant aggregation as the key design choice.
2. A controlled comparison of four aggregation strategies (RQ1) with a CLIP-based fidelity/consistency evaluation, error bars, and a significance test, on a from-scratch crowd→image pipeline (Figure 1).

3. A diagnosis and mitigation of the CLIP modality gap for crowd conditioning, with the failed inference-time fixes reported as negative results.
4. Evidence that the aggregation finding reproduces across two generative backbones (diffusion and GAN), indicating it is a property of aggregation rather than of one generator (RQ2).

1.1 Novelty

Unlike conventional conditional-diffusion studies, this work targets three things at once:

- Many-to-one conditioning — a *set* of hundreds of inputs, not a single prompt.
- Permutation-invariant aggregation as the object of study — the aggregator, not the generator, is the variable we vary and measure.
- Crowd-driven generation — noisy, heterogeneous, collective input rather than a curated caption.

To our knowledge, existing conditional image generation assumes a single conditioning signal, whereas this work studies conditioning on an unordered set of heterogeneous audience responses. In other words, the generator is *infrastructure*; the aggregator is the object of study.

2. Problem formulation and research questions

Given a crowd of responses $X = \{x_1, x_2, \dots, x_N\}$ (words / emojis, order-free), we seek an aggregation function A and a conditional generator G such that

$$A : X \rightarrow c \in \mathbb{R}^d \text{ and } I = G(z, c) \text{ maximize theme-fidelity of the image } I,$$

where z is a latent noise sample from a standard Gaussian prior, and A must be permutation-invariant ($A(X) = A(\pi(X))$ for any permutation π) and robust to off-theme noise in X . The generator G and the encoder that produces the per-item embeddings are fixed; A is the object of study. Two questions follow:

- **RQ1 (primary).** Among permutation-invariant choices of A (mean, cluster-centroid, Deep-Sets, attention-PMA), which best turns a noisy crowd into an on-theme image — and how does the answer change as crowd diversity (fraction of off-theme responses) increases?
- **RQ2.** Do the observed gains arise from the aggregation mechanism itself, or are they specific to the diffusion generator? (We test the same A 's under a GAN G .)

3. Related work

Diffusion models. Denoising diffusion probabilistic models (Ho et al., 2020) generate by learning to reverse a gradual noising process; latent diffusion (Rombach et al., 2022) moves this into a VAE latent space for efficiency, which we adopt. Classifier-free guidance (Ho & Salimans, 2022) trades diversity for fidelity by extrapolating between conditional and unconditional predictions; we use it as the control knob.

Set functions. Because a crowd is an unordered set, aggregation must be permutation-invariant. Deep Sets (Zaheer et al., 2017) realize this as $\rho(\sum \phi(\cdot))$; the Set Transformer's pooling-by-multihead-attention, PMA (Lee et al., 2019), learns a query that attends over the set. These are our two learned aggregators; mean- and cluster-centroid pooling are the heuristic baselines.

Vision-language embeddings. CLIP (Radford et al., 2021) embeds images and text into a shared space, which is what lets a crowd of *text* condition an *image* generator; it also gives us a training-free

evaluation metric. We rely on, and must contend with, CLIP's modality gap — image and text embeddings occupy different regions of that shared space.

GANs. For the baseline we use a conditional GAN with a projection discriminator (Miyato & Koyama, 2018) and the hinge loss, in the same latent space, for a controlled comparison.

4. Method

4.1 System overview

The pipeline is shown in Figure 1. A crowd of responses is embedded by CLIP into a set of vectors; an aggregator collapses that set into one conditioning vector $\mathbf{c}$; a conditional generator (diffusion or GAN) produces a latent from $\mathbf{c}$; and the frozen VAE decodes it to an image. The VAE is trained once and frozen, and aggregation is applied only at inference, which makes the aggregator a swappable study variable — every experiment changes only the aggregation step.

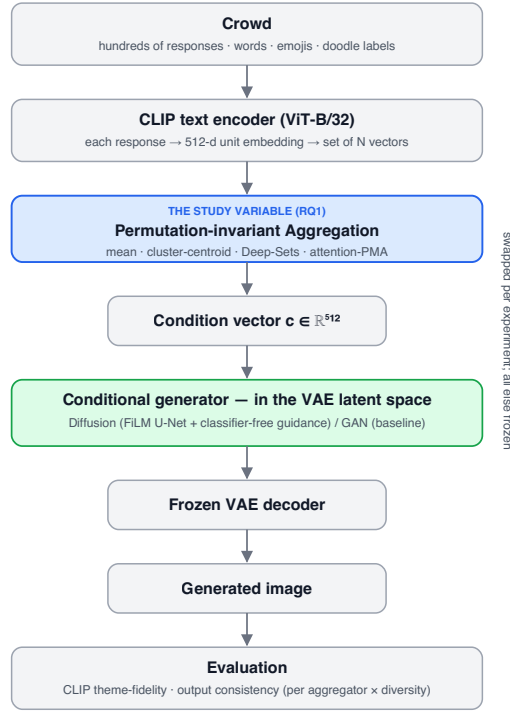


Figure 1. The crowd → image pipeline. Everything except the highlighted aggregation block is held fixed across experiments, isolating aggregation as the variable under study (RQ1).

4.2 Crowd simulator

Public datasets of large, theme-labelled crowd responses are not available, so we use a controlled simulator (which also lets us vary the noise level precisely). It produces crowds for five themes (*paradise*, *fire*, *love*, *night*, *hope*) over word, emoji, and doodle-label vocabularies (all treated as short text), with a `diversity` knob controlling the fraction of off-theme noise (0 = coherent, 1 = pure noise). This yields labelled, reproducible crowds at any noise level — the independent variable for the robustness study.

4.3 Encoders

Responses are embedded with CLIP (ViT-B/32)'s text encoder into 512-d unit vectors; images are embedded with its image encoder. CLIP's approximately aligned shared embedding space is what allows *text* to steer an *image* generator — though, as §6.2 shows, image and text embeddings are aligned but not identical, which we must contend with.

4.4 Aggregation strategies (the study variable)

All four map a set of embeddings to one unit vector:

- mean-pool — `normalize(mean_i e_i)`; the naïve baseline (averages everything).
- cluster-centroid — k-means, take the dominant cluster's centroid (robust to scattered noise).
- Deep-Sets — $\rho(\sum_i \phi(e_i))$, learned.
- attention-PMA — a learned seed query attends over the set, learned.

The two learned aggregators are trained with weak supervision from theme labels (§4.8).

4.5 Stage 1 — Variational Autoencoder

A convolutional β -VAE maps 64×64 RGB to a spatial latent. Latent size is the dominant control on reconstruction quality; after ablation we use a `[4,16,16]` latent ($4 \times$ downsampling), $\beta = 0.1$, and an LPIPS perceptual loss for crisp reconstructions. Both generators operate in this frozen latent.

4.6 Stage 2 — Conditional latent diffusion

A FiLM-conditioned U-Net predicts the noise $\epsilon(z_t, t, c)$ with a cosine schedule and **classifier-free guidance** (condition-dropout at training; guided DDIM sampling with scale $w = 3$). Latents are scaled to $\sim$ unit variance before diffusion.

4.7 GAN baseline

A conditional GAN in the *same* latent with the *same* CLIP conditioning: a generator `(noise  $\oplus$  c)  $\rightarrow$  latent` and a projection discriminator, trained with the hinge loss. Inference is identical to the diffusion path, enabling a fair backbone comparison (RQ2).

4.8 Training the learned aggregators

Weakly supervised using the crowd's known theme label as the target (no generator in the loop): given a noisy crowd's embeddings, the aggregator is trained to output a vector close to the theme's clean embedding, `loss = 1 - cos(agg(crowd), CLIP_text("a {theme} scene"))`. This directly rewards robust theme extraction.

4.9 Design choices at a glance

Component	Choice
Conditioning encoder	CLIP ViT-B/32 (shared image–text space)
Latent space	Convolutional β -VAE, $[4, 16, 16]$, $\beta = 0.1$, + LPIPS
Primary generator	Conditional latent diffusion (FiLM U-Net, cosine schedule, DDIM)
Guidance	Classifier-free guidance, $w = 3$
Aggregators	mean · cluster-centroid · Deep-Sets · attention-PMA
Baseline generator	Conditional GAN (projection discriminator, hinge loss)
Resolution	64×64 RGB
Dataset	WikiArt (streamed subset)

5. Experiments

5.1 Setup

Data. We train on WikiArt ([hugging/wikiart](https://huggingface.co/datasets/huggingface/wikiart)), streamed at 64×64 and normalized to $[-1, 1]$. WikiArt provides diverse artistic styles and genres while remaining small enough to train diffusion models from scratch on a single GPU; its genre labels also give a text signal we exploit to fix the modality gap (§6.3). An artistic domain is a natural fit — coherent *mood/style* matters more than photographic accuracy.

Training & hardware. VAE: 5k images, 40 epochs. Diffusion (final, "v2"): mixed image + WikiArt genre-label text conditioning, 15k images, base-128 U-Net, 120 epochs. GAN: same data/conditioning. All on Google Colab (T4 / A100), monitored with **Weights & Biases**; checkpoints persisted to Google Drive and the **HuggingFace Hub**. Learned aggregators are trained at crowd size $n = 300$ to match evaluation.

Run	Parameters	GPU	~Wall-clock (incl. one-time streaming)
VAE	4.3 M	T4	~10–15 min
Diffusion (v2)	27.3 M (U-Net)	A100	~35 min (~20 min streaming + ~7 min training on pre-encoded latents)
GAN	6.7 M (G) / 2.9 M (D)	A100	~30–35 min
Deep-Sets / attention-PMA	0.39 M / 1.84 M	T4	~3 min
RQ1 evaluation (per pass)	—	T4	~15–25 min

(GPU training itself is fast — CLIP and the VAE are frozen and latents/conditions are pre-encoded once, so each diffusion/GAN epoch is cheap; single-threaded WikiArt streaming is usually the dominant wall-clock cost. Beyond these final runs, the full study — a β /latent-size ablation, the modality-gap diagnosis with two abandoned inference-time fixes (gap correction, guidance sweep), a first diffusion run superseded by the text-conditioned retrain, the GAN baseline, aggregator training, and several 24-

repeat evaluation passes across both backbones — spanned dozens of Colab executions, roughly 6–10 hours across T4 and A100 sessions. Per-run durations are logged in *Weights & Biases*.)

5.2 Metrics

- Theme-fidelity = `cos(CLIP_img(generated), CLIP_text("a {theme} scene"))` — is the image on-theme? CLIP is trained so that matching image–text pairs have high cosine similarity, so this cosine is a reasonable training-free proxy for "does the image depict the theme."
- Consistency = mean pairwise CLIP-similarity among images generated from *independent* crowds of the same theme — does the aggregator map noisy crowds to a stable target?

Each (aggregator, diversity) cell reports mean $\pm$ standard error over `repeats  $\times$  themes` = $24 \times 5 = 120$ generated images, with error bars. We deliberately avoid FID/IS: there is no paired ground truth for a crowd, so distributional-similarity scores would not be meaningful here.

6. Results

6.0 Ablation summary

Experiment	Observation
KL weight $\beta \uparrow$	smoother but blurrier reconstructions
Latent size $\uparrow$ ([4,8,8] $\rightarrow$ [4,16,16])	biggest single improvement to reconstruction
LPIPS perceptual loss	helps only <i>once</i> the latent has capacity
Guidance scale $\uparrow$ (CFG $w = 3 \rightarrow 10$)	amplifies but cannot fix weak text conditioning
Additive modality-gap correction	fails — collapses themes toward the mean image
Text-conditioned retrain	substantially mitigates the modality mismatch (decisive RQ1 separation)

6.1 VAE reconstruction (β / latent ablation)

The latent bottleneck, not the loss, was the dominant limiter. A [4,8,8] latent gave blurry colour-blob reconstructions regardless of β or LPIPS; enlarging to [4,16,16] ($4\times$ more capacity) produced crisp, recognizable reconstructions. β trades sharpness against latent regularity ($\beta = 1.0$ over-smoothed; $\beta = 0.1$ was sharpest). LPIPS helped only once the latent had capacity.



Figure 2. VAE reconstructions on WikiArt (top: input paintings; bottom: reconstructions from the [4,16,16] latent). Structure and palette are preserved — a good latent for the generator.

6.2 Diffusion training and the modality gap

The diffusion model converged cleanly (ϵ -prediction $\text{MSE} \approx 1.1 \rightarrow 0.4$) and, conditioned on an image's *own* CLIP image embedding, produced theme-matched generations — confirming the generator responds to conditioning. But conditioned on a crowd's CLIP text embedding, theme control was weak and inconsistent (e.g. `night` rendered *light*; Figure 3). Because image-embedding conditioning worked and text-embedding conditioning did not, we attribute this to the CLIP modality gap (image and text embeddings occupy different regions of the shared space; the model, trained only on image-embedding conditions, sees text conditions as out-of-distribution).

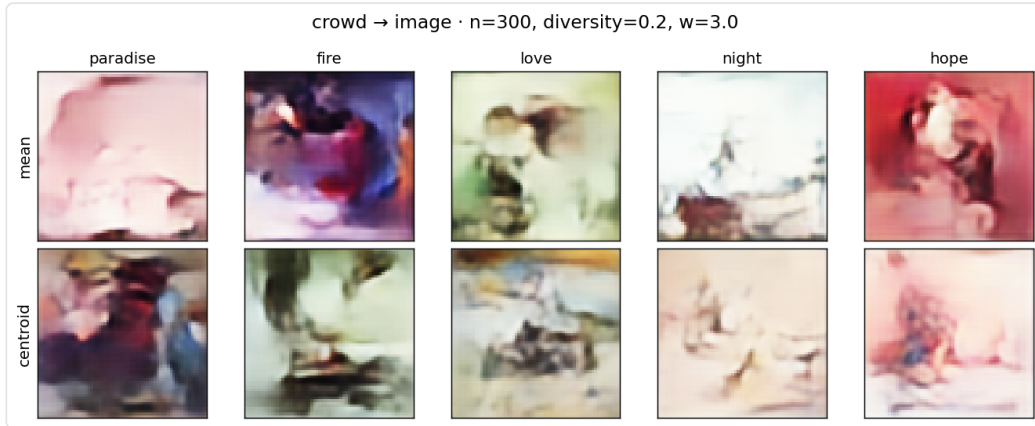


Figure 3. Weak text conditioning before the retrain: crowd $\rightarrow$ image on the image-only diffusion (mean/centroid $\times$ five themes). Themes are barely differentiated and `night` renders *light* — the visible symptom of the modality gap.

Negative results (reported). Two inference-time fixes failed: (i) an additive modality-gap correction collapsed all themes toward the image-cloud mean (a washed-out "average painting"); (ii) a guidance sweep ($w = 3 \rightarrow 10$) amplified the weak signal but never corrected it. Both are documented rather than hidden.

6.3 Text-conditioned retrain (before / after)

We repair the gap at training time: condition on a mix of each image's CLIP image embedding and the CLIP text embedding of its WikiArt genre label (`p_text = 0.5`), teaching the model that text embeddings are valid conditions. The genre labels resolve to content-oriented names (landscape, cityscape, portrait, ...), which provide coarse semantic supervision that partially transfers to our scene/mood themes. This turned an *indistinguishable* aggregator comparison into a decisive one: on the image-only model all four aggregators were statistically tangled (~ 0.18 , overlapping error bars), whereas after the retrain a clear learned-vs-heuristic separation emerges (§6.4), and text-conditioned samples become visibly structured and theme-appropriate (§6.5).

6.4 RQ1 — the four-way aggregation comparison

On the retrained model, with 24 repeats and error bars, theme-fidelity separates into two clear tiers with non-overlapping error bars:

aggregator	div 0.1	div 0.4	div 0.7
mean	0.162 $\pm$ 0.002	0.161 $\pm$ 0.002	0.162 $\pm$ 0.002
centroid	0.163 $\pm$ 0.002	0.162 $\pm$ 0.002	0.162 $\pm$ 0.002
deepsets	0.196 $\pm$ 0.003	0.195 $\pm$ 0.003	0.194 $\pm$ 0.003
attention	0.197 $\pm$ 0.003	0.194 $\pm$ 0.003	0.196 $\pm$ 0.003

Learned pooling (attention $\approx$ Deep-Sets, ~ 0.195) $\gg$ heuristic pooling (centroid $\approx$ mean, ~ 0.162), holding flat across all diversity levels — the learned aggregators remain on-theme as the crowd gets noisier. A Welch's t-test on the pooled per-image fidelities confirms the separation: learned 0.195 vs heuristic 0.162, difference 0.034, 95% CI [0.031, 0.036], $t = 23.8$, $p \approx 3 \times 10^{-103}$ ($n = 720$ per tier). Within tiers, attention vs Deep-Sets and centroid vs mean are not statistically distinguishable. Interpreted plainly: **which aggregation function is used matters far more** than which of the two heuristics, or (as §6.6 shows) which generator, is chosen — the conditioning representation is the dominant lever.

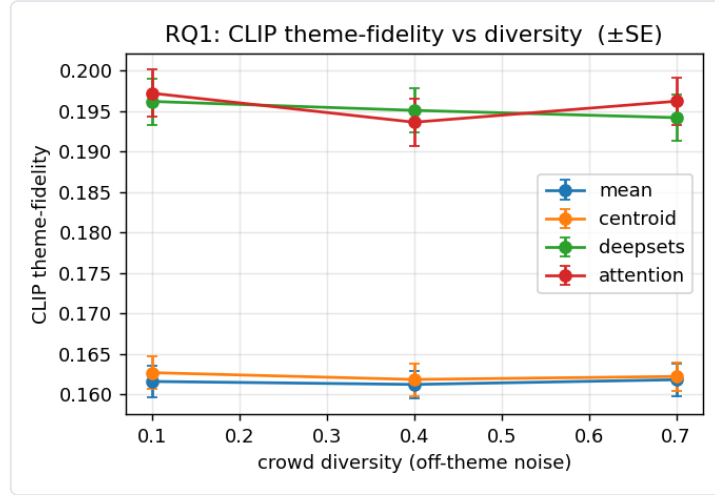


Figure 4. RQ1 — CLIP theme-fidelity vs crowd diversity (mean $\pm$ SE). Two non-overlapping tiers: learned pooling (attention $\approx$ Deep-Sets) $\gg$ heuristic pooling (centroid $\approx$ mean), across all diversity levels.

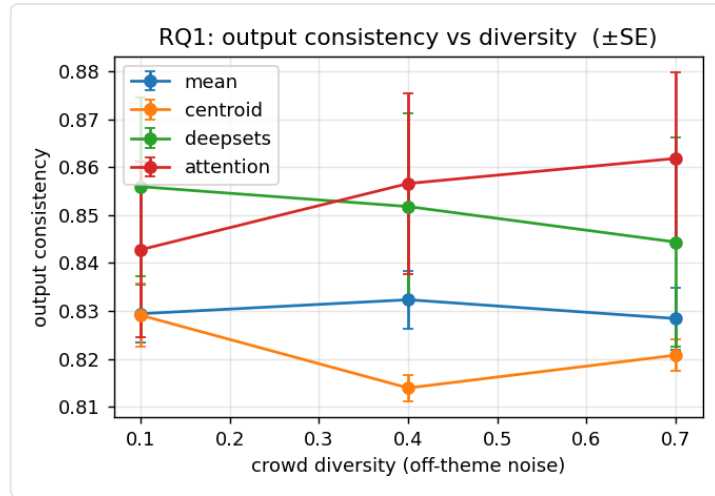


Figure 5. RQ1 — output consistency vs diversity (mean $\pm$ SE): learned $\geq$ heuristic; attention most consistent at maximum noise.

Note on rigour. An initial low-repeat run had suggested finer orderings ("centroid > mean", "mean collapses under noise"); these did not survive 24 repeats (overlapping bars). We report the two-tier claim, which is what the data supports.

6.5 Qualitative results

The qualitative grids corroborate the numbers. In the themes grid (Figure 6), mean/centroid produce pale, washed-out images while Deep-Sets/attention produce coherent, painterly, theme-appropriate scenes (paradise → green landscape, fire → dark harbour, night → genuinely dark). In the diversity grid (Figure 7), the learned aggregators hold a consistent green-paradise landscape across diversity 0.1 → 0.7 while the heuristics drift.

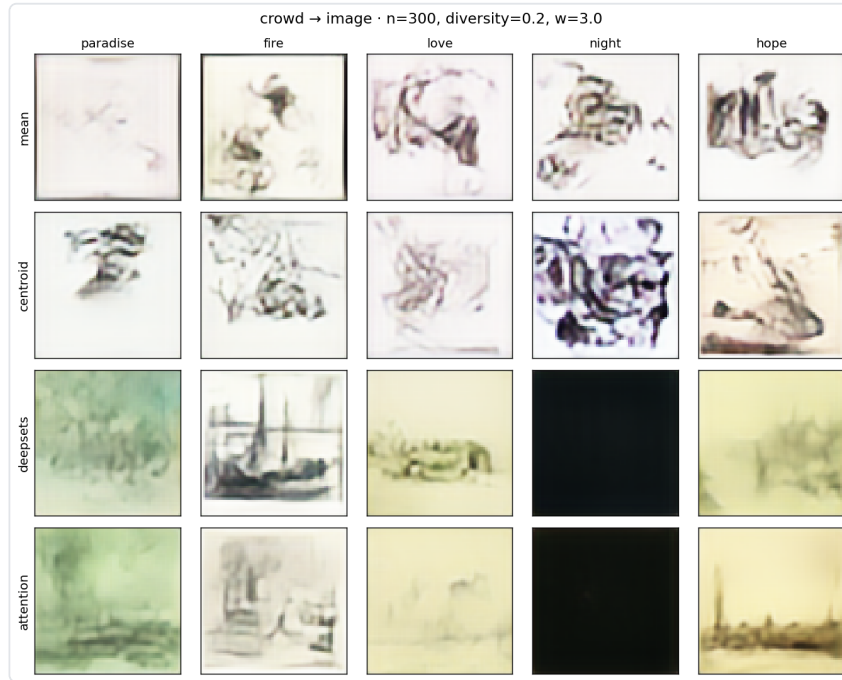


Figure 6. Qualitative crowd→image on the retrained model — four aggregators (rows) × five themes (columns). Learned aggregators produce coherent, theme-appropriate scenes; heuristics are pale and washed-out.

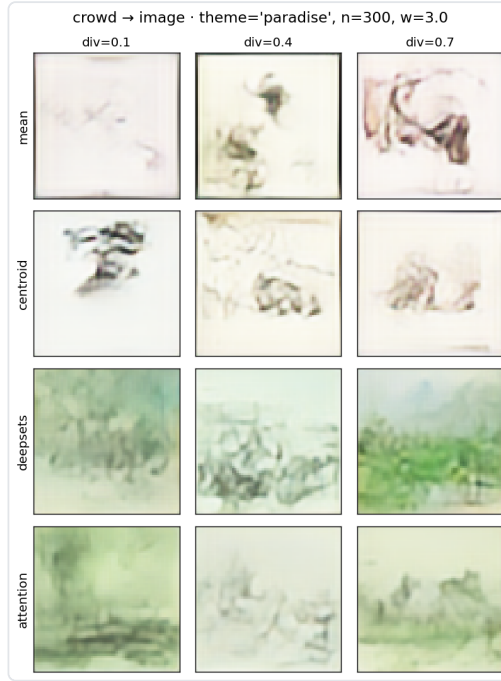


Figure 7. Robustness — same theme (paradise), rising off-theme noise (columns). The learned aggregators hold a consistent landscape across diversity; the heuristics drift.

6.6 RQ2 — is the effect from aggregation, or from diffusion?

RQ2 asks whether the RQ1 gains arise from the aggregation mechanism itself or are specific to the diffusion generator. We answer it by swapping in the GAN and re-running the identical evaluation. The conditional GAN trained stably (**no mode collapse**) and produced coherent, soft painterly samples (Figure 8). Crucially, the learned $\gg$ heuristic separation reproduces on the GAN (Figure 9): non-overlapping error bars (Welch's t-test: difference 0.015, 95% CI [0.012, 0.019], $p \approx 8 \times 10^{-15}$). Because the same ordering appears under a completely different generator, the effect is a property of the aggregation, not of the diffusion model.

metric	diffusion_v2	GAN
learned-aggregator fidelity	~ 0.195	~ 0.190
heuristic-aggregator fidelity	~ 0.162	~ 0.176
output consistency	$\sim 0.81\text{--}0.86$	$\sim 0.78\text{--}0.83$

The two backbones are comparable on fidelity (diffusion edges the learned tier; the GAN edges the heuristic tier), while diffusion is more consistent. At this scale both are viable; diffusion is the more reliable generator.

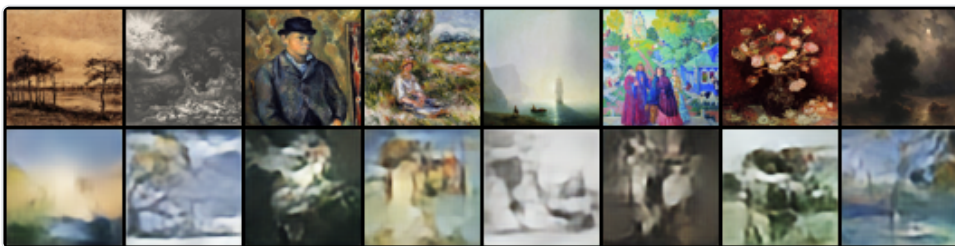


Figure 8. Conditional GAN generations (bottom) from label-text conditioning vs real paintings (top) — coherent and varied, no mode collapse.

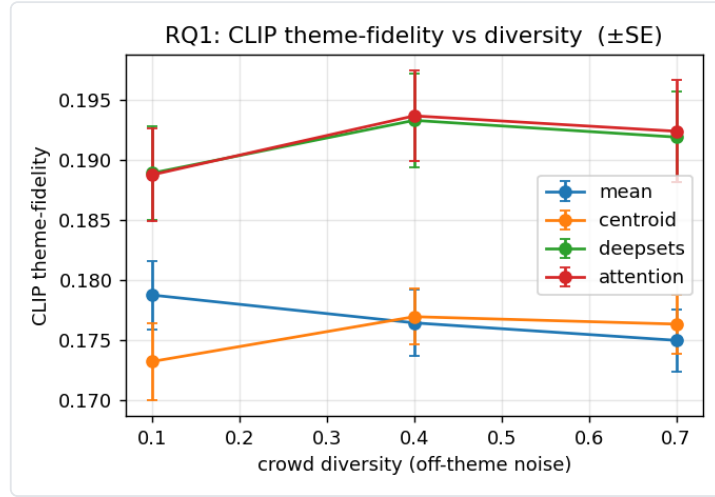


Figure 9. RQ1 on the GAN backbone (theme-fidelity vs diversity, $\pm$ SE). The learned $\gg$ heuristic separation reproduces — the aggregation effect holds across two generative backbones.

7. Discussion and limitations

Why do learned aggregators win? Mean- and centroid-pooling apply *fixed* rules: mean weights every response equally (so off-theme responses drag the condition toward mush), and the centroid commits to one cluster. The learned aggregators can instead adapt to the set. We hypothesize they learn behaviours a fixed rule cannot: (i) noise rejection — down-weighting responses far from the emerging theme; (ii) theme extraction — mapping the set to the dominant semantic direction; and (iii) importance weighting — giving informative responses more influence. Attention-PMA is architecturally suited to this via its query-key attention; Deep-Sets through the learned ϕ / ρ maps. We do not isolate these mechanisms directly, but they are consistent with the learned aggregators' advantage *holding* rather than shrinking as noise rises.

Limitations.

- **Teaching-to-the-test (important).** The learned aggregators are trained to output $\approx \text{CLIP_text}(\text{theme})$, and the fidelity metric is $\cos(\text{image}, \text{CLIP_text}(\text{theme}))$ — so their objective is partly aligned with the metric. Their advantage is real (they are also more robust and more consistent, which the objective does not directly reward), but its *magnitude* is inflated by this alignment; the heuristics receive no such supervision. A fully arms-length metric (human rating, or a different theme-similarity measure) is the rigorous follow-up.
- **Absolute quality / scale.** Models are small and trained from scratch at 64×64 . Absolute CLIP fidelity is modest (~ 0.16 – 0.20 vs ~ 0.28 – 0.35 for well-matched pairs); the relative aggregator comparison is the result, not the absolute image quality.
- **Genre $\neq$ mood.** The text conditioning is learned from WikiArt *genre* labels (content categories), not the mood themes we test; transfer is imperfect.
- **Simulated crowds.** Results use a crowd simulator, not human crowds. The simulator also treats responses as independent — it does not model social influence (e.g. later responses echoing earlier ones), which would change a real crowd's diversity distribution.

8. Future work

We group directions from near-term *research* to eventual *application*.

- **Research.** (i) Arms-length evaluation — a human study and a held-out similarity metric to remove the teaching-to-the-test bias; (ii) multimodal conditioning — extend beyond words/emojis to sketches (doodle images) and audio (cheers, music) in the same shared space.
- **Engineering.** (iii) Streaming / online aggregation — update the condition as responses arrive rather than batching a fixed crowd; (iv) higher-capacity generation — larger-resolution latent diffusion for crisper output, and video diffusion to animate a crowd's evolving mood.
- **Application.** (v) Real audience deployment — drive generation from live crowds at events, closing the loop with the human ratings from (i).

9. Conclusion

We built a complete crowd → image system and used it to answer a focused design question: learned permutation-invariant aggregation significantly outperforms heuristic pooling for turning a noisy crowd into a coherent image — robustly across crowd diversity, and the ordering reproduces under a GAN generator, with a large, highly significant effect. Reaching this required diagnosing and mitigating the CLIP modality gap, and the negative results and limitations are reported alongside the positive ones. Within the scope of our experiments, the results suggest that for many-to-one conditioning problems, learning the conditioning representation is a more impactful lever than changing the underlying generative backbone.

Appendix — reproducibility

Code on branch `feature/crowd-driven-visual-generation`; notebooks `01–07` (Colab), each auto-locating checkpoints from Google Drive or the HuggingFace Hub. Training logs to Weights & Biases; artifacts (checkpoints, caches) are git-ignored and hosted on Drive + HF (`crowdgen-vae` , `crowdgen-diffusion(-v2)` , `crowdgen-gan`). Key configs: VAE `[4,16,16]` / $\beta=0.1$ /LPIPS; diffusion FiLM-U-Net base-128, cosine schedule, DDIM-50, $w=3$; aggregators trained at $n=300$; evaluation 24 repeats $\times$ 5 themes, mean $\pm$ SE with a Welch's t-test on pooled per-image fidelities.

References

1. **Ho, J., et al. (2020).** *Denoising Diffusion Probabilistic Models*. NeurIPS 2020.
2. **Rombach, R., et al. (2022).** *High-Resolution Image Synthesis with Latent Diffusion Models*. CVPR 2022.
3. **Ho, J., & Salimans, T. (2022).** *Classifier-Free Diffusion Guidance*. NeurIPS Workshop 2022.
4. **Zaheer, M., et al. (2017).** *Deep Sets*. NeurIPS 2017.
5. **Lee, J., et al. (2019).** *Set Transformer: A Framework for Attention-based Permutation-Invariant Neural Networks*. ICML 2019.
6. **Radford, A., et al. (2021).** *Learning Transferable Visual Models from Natural Language Supervision (CLIP)*. ICML 2021.
7. **Miyato, T., & Koyama, M. (2018).** *cGANs with Projection Discriminator*. ICLR 2018.